GOOD AND BAD REDUCTION: EXAMPLES

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An elliptic curve over $\mathbb{Q}_p$ has good, multiplicative or additive reduction, and multiplicative reduction is split or nonsplit [Sil09, §VII.5]. The problems below run through concrete curves of each kind: Problem 1.8 reads the type off an equation, Problem 2.2 counts the rational points of the singular reduction, and Problem 3.1 changes the type by twisting the curve or extending the field. Problem 4.1 ($\star$) goes further: the curves Frey attached to solutions of $A + B = C$ have no additive reduction at all. Each problem is preceded by the definitions and results it uses.

Standing notation.

- A prime p, with $v = v_p$ its valuation on $\mathbb{Q}_p$, extended to $\overline{\mathbb{Q}}_p$.
- An elliptic curve E over $\mathbb{Q}_p$, given by a Weierstrass equation with coefficients in $\mathbb{Z}_p$, with discriminant Δ and invariants c_4, c_6 [Sil09, §III.1].
- When the equation is minimal, E_p is its reduction modulo p, a Weierstrass cubic over $\mathbb{F}_p$.
- We write
$$a_p \coloneqq p + 1 - \#E_p(\mathbb{F}_p),$$
counting the singular point too.

How to use this. Allow about 15 minutes for each of Problems 1.8, 2.2 and 3.1. Problem 4.1 is for home. The one `Magma` block, in Problem 1.8, runs in seconds in the free online calculator. Everything else is by hand.

1. Reading off the type

The following definition and results hold for every prime p, and are used throughout. In them, K is $\mathbb{Q}_p$ or a finite extension of it, with normalized valuation v_K, ring of integers $\mathcal{O}_K$ and residue field k, and E is an elliptic curve over K given by a Weierstrass equation with coefficients in $\mathcal{O}_K$.

Definition 1.1 (Reduction types [Sil09, §VII.5])**.** Suppose that the equation is minimal, and consider its reduction, a Weierstrass cubic over k. Then E has good reduction if the reduction is nonsingular, multiplicative reduction if the reduction has a node, and additive reduction if it has a cusp. Multiplicative reduction is split if the slopes of the tangent lines at the node lie in k, and nonsplit otherwise.

Proposition 1.2 (Discriminant and c_4 [Sil09, §III.1])**.** *For an equation $y^2 = x^3 + a_2x^2 + a_4x + a_6$ (that is, $a_1 = a_3 = 0$),*

$$\Delta = -16\big(4a_2^3a_6 - a_2^2a_4^2 - 18a_2a_4a_6 + 4a_4^3 + 27a_6^2\big), \qquad c_4 = 16\big(a_2^2 - 3a_4\big), \tag{1.1}$$

which follow from the formulas of.

Proposition 1.3 (Minimal equations [Sil09, Remark VII.1.1])**.** *Any other Weierstrass equation for E has discriminant $u^{-12}\Delta$ and invariant $u^{-4}c_4$ for some $u \in K^\times$. So if $v_K(\Delta) < 12$ or $v_K(c_4) < 4$, the equation is minimal.*

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Proposition 1.4 (Singular Weierstrass cubics [Sil09, Proposition III.1.4(a)]). *A Weierstrass cubic over a field with $\Delta = 0$ has exactly one singular point over an algebraic closure. It is a node, with two distinct tangent lines, if $c_4 \neq 0$, and a cusp, with one tangent line, if $c_4 = 0$.*

Proposition 1.5 (Criteria for the reduction type [Sil09, Proposition VII.5.1]). *Suppose that the equation is minimal. Then E has good reduction if $v_K(\Delta) = 0$, multiplicative reduction if $v_K(\Delta) > 0$ and $v_K(c_4) = 0$, and additive reduction if $v_K(\Delta) > 0$ and $v_K(c_4) > 0$.*

Proposition 1.6 (Extending the field [Sil09, Proposition VII.5.4(b)]). *Let L be a finite extension of K with ramification index e, so that $v_L = e\, v_K$ on K. If E has good, respectively multiplicative, reduction, then so does E_L.*

Proposition 1.7 (Potential good reduction [Sil09, Proposition VII.5.5]). *There is a finite extension L of K such that E_L has good reduction if and only if $j(E) \in \mathcal{O}_K$.*

Problem 1.8 (Reading off the type).
Let $p \geqslant 5$, and let $\varepsilon \in \mathbb{Z}$ be a non-square modulo p. For each equation, decide whether it is minimal at p, and find the reduction type of the curve over $\mathbb{Q}_p$: good, split multiplicative, nonsplit multiplicative, or additive.
(1) $y^2 = x^3 + x^2 + p$.
(2) $y^2 = x^3 + \varepsilon x^2 + p$.
(3) $y^2 = x^3 + p$.
(4) $y^2 = x^3 + p^6$.
(5) $y^2 = x^3 + x^2 + p^{12}$.
(6) Here is Magma's opinion at $p = 7$ and $\varepsilon = 3$. For each curve it prints the valuation of the minimal discriminant, the Kodaira symbol of the reduction (I_0 for good reduction, I_n with $n \geqslant 1$ for multiplicative, anything else for additive), and the polynomial $1 - a_7 T$, or $1 - a_7 T + 7T^2$ in the case of good reduction.

```
R<T> := PolynomialRing(Integers());
p := 7; eps := 3;                          // 3 is not a square modulo 7
Info := function(a)
  E := EllipticCurve(a); L := LocalInformation(E, p);
  return <L[2], L[5], R!EulerFactor(E, p)>;
end function;
Info([0, 1, 0, 0, p]);                     // <1, I1, -T + 1>
Info([0, eps, 0, 0, p]);                   // <1, I1, T + 1>
Info([0, 0, 0, 0, p]);                     // <2, II, 1>
Info([0, 0, 0, 0, p^6]);                   // <0, I0, 7*T^2 + 4*T + 1>
Info([0, 1, 0, 0, p^12]);                  // <12, I12, -T + 1>
```

Check that it agrees with your answers. Note the values of a_7 at the curves with bad reduction. Problem 2.2 explains them.

2. Counting rational points on the singular reduction

Definition 2.1 (The nonsingular locus). When the equation of E is minimal, the nonsingular locus E_p^{ns} is E_p minus its singular point, if it has one.

Problem 2.2 (Counting with lines through the singular point)**.**
Let E be an elliptic curve over $\mathbb{Q}_p$ with bad reduction, and let S be the singular point of its reduction E_p.

(1) Show that $S \in E_p(\mathbb{F}_p)$, and that $S \neq O$.[a]

(2) Show that a change of variables over $\mathbb{F}_p$ moves S to $(0,0)$ and puts E_p in the form

$$y^2 + a_1xy = x^3 + a_2x^2,$$

and that the tangent lines at S are the lines $y = \alpha x$ with $g(\alpha) = 0$, where $g(t) \coloneqq t^2 + a_1t - a_2$.

(3) Let $t \in \mathbb{F}_p$. Show that the line $y = tx$ meets E_p at S and, if $g(t) \neq 0$, at exactly one more $\mathbb{F}_p$-rational point $P_t = \big(g(t), t\,g(t)\big)$, and nowhere else in the affine plane. Show that $t \mapsto P_t$ is a bijection from $\{t \in \mathbb{F}_p : g(t) \neq 0\}$ onto $E_p^{\mathrm{ns}}(\mathbb{F}_p) - \{O\}$.

(4) Deduce that $\#E_p^{\mathrm{ns}}(\mathbb{F}_p) = p + 1 - r$, where r is the number of roots of g in $\mathbb{F}_p$, and hence that

$$a_p = \begin{cases} 1 & \text{for split multiplicative reduction,} \\ -1 & \text{for nonsplit multiplicative reduction,} \\ 0 & \text{for additive reduction.} \end{cases}$$

Compare with the output in Problem 1.8(6).

(5) Check (4) by hand for $y^2 = x^3 + x^2$ and $y^2 = x^3 + 2x^2$ over $\mathbb{F}_5$.

[a] *Hint:* what does the Frobenius automorphism of $\overline{\mathbb{F}}_p$ do to a singular point?

3. Changing the curve, changing the field

Problem 3.1 (Twists and extensions)**.**
Let $p \geqslant 5$. For $d \in {\mathbb{Q}_p}^\times$ and $E\colon y^2 = x^3 + a_2x^2 + a_4x + a_6$, the quadratic twist of E by d is

$$E^d\colon y^2 = x^3 + da_2x^2 + d^2a_4x + d^3a_6.$$

(1) Show that $(x, y) \mapsto (dx, d^{3/2}y)$ is an isomorphism $E \to E^d$ over $\mathbb{Q}_p(\sqrt{d})$, and that $\Delta(E^d) = d^6\Delta(E)$, $c_4(E^d) = d^2c_4(E)$ and $j(E^d) = j(E)$.

From now on, $E\colon y^2 = x^3 + x^2 + p$ is the curve of Problem 1.8(1), with split multiplicative reduction.

(2) Let $\varepsilon \in \mathbb{Z}$ be a non-square modulo p. Show that E^ε has nonsplit multiplicative reduction, and that $a_p(E^\varepsilon) = -a_p(E)$. Since $\mathbb{Q}_p(\sqrt{\varepsilon})$ is an unramified extension of $\mathbb{Q}_p$, what does this say about which features of the reduction are preserved by unramified extensions?

(3) Show that $E^p\colon y^2 = x^3 + px^2 + p^4$ has additive reduction. Over which quadratic extension of $\mathbb{Q}_p$ does it acquire multiplicative reduction? Show that it has good reduction over no finite extension of $\mathbb{Q}_p$.

(4) Let $E_3\colon y^2 = x^3 + p$, and let L be a finite extension of $\mathbb{Q}_p$ with ramification index e. Show that E_3 has good reduction over L if and only if $6 \mid e$.[a]

(5) Show that $y^2 = x^3 + px$ has additive reduction, and good reduction over L as in (4) if and only if $4 \mid e$.

[a] *Hint:* for "if", write $p = \varpi^e w$ with ϖ a uniformizer of L and w a unit.

4. ⋆ Frey curves

This section is [Sil09, Exercise 8.23], with the conductor of [Mil20, Aside II.3.5]. The LMFDB [LMF25] is the answer key.

Problem 4.1 (Frey curves are semistable)**.**
An elliptic curve over $\mathbb{Q}$ is semistable if it has good or multiplicative reduction at every prime. Let A, B, C be nonzero integers with $A + B = C$ and $\gcd(A, B, C) = 1$. Let

$$E\colon y^2 = x(x + A)(x - B).$$

(1) Show that $\Delta = 16A^2B^2C^2$ and $c_4 = 16(A^2 + AB + B^2)$.
(2) Let p be an odd prime dividing ABC. Show that the equation is minimal at p and that E has multiplicative reduction at p. Show that the reduction is split if and only if $-B$, A or B is a square modulo p, according as p divides A, B or C.
(3) Suppose that $A \equiv 0 \pmod{16}$ and $B \equiv 3 \pmod 4$. Show that the substitution $x = 4x'$, $y = 8y' + 4x'$ gives an equation with coefficients in $\mathbb{Z}$,

$$y'^2 + x'y' = x'^3 + \frac{A - B - 1}{4}\,x'^2 - \frac{AB}{16}\,x',$$

with discriminant $(ABC)^2/2^8$ and $c_4 = A^2 + AB + B^2$. Deduce that E has good reduction at 2 if $v_2(A) = 4$, and multiplicative reduction at 2 if $v_2(A) \geqslant 5$. So E is semistable.
(4) The conductor of E is $N = \prod_p p^{f_p}$, where $f_p = 0$ if E has good reduction at p, $f_p = 1$ if it has multiplicative reduction, and $f_p \geqslant 2$ if it has additive reduction. Under the hypotheses of part (3), compute N in terms of A, B, C. Find the curves for $(A, B, C) = (16, -1, 15)$ and $(32, -5, 27)$ in the LMFDB, and check their conductors and their split and nonsplit primes against your answers.
(5) Suppose that $a^\ell + b^\ell = c^\ell$ with $\ell \geqslant 5$ prime and a, b, c nonzero coprime integers. Show that, after reordering and changing signs, $(A, B, C) = (a^\ell, b^\ell, c^\ell)$ satisfies the hypotheses of part (3), and that the resulting curve has conductor $\prod_{p|abc} p$ and minimal discriminant $(abc)^{2\ell}/2^8$.

References

[LMF25] The LMFDB Collaboration, *The L-functions and modular forms database*, 2025, https://www.lmfdb.org. [Online; accessed 21 July 2025]. ↑

[Mil20] James S. Milne, *Elliptic curves*, Second, World Scientific Publishing Co. Pte. Ltd., Hackensack, NJ, 2020. ↑

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑